National identity cards

Egypt and Germany

Design, materials, printing and security

Egypt



Germany



Educational examples from the lecture and official German guide

Design and data layout

Egypt



Arabic data and national motifs form the visual identity.

Germany



Multilingual labels and national symbols form the visual identity.

Both designs combine identity information with security patterns.

[L1] pp. 13–19 [B] p. 2

Card materials

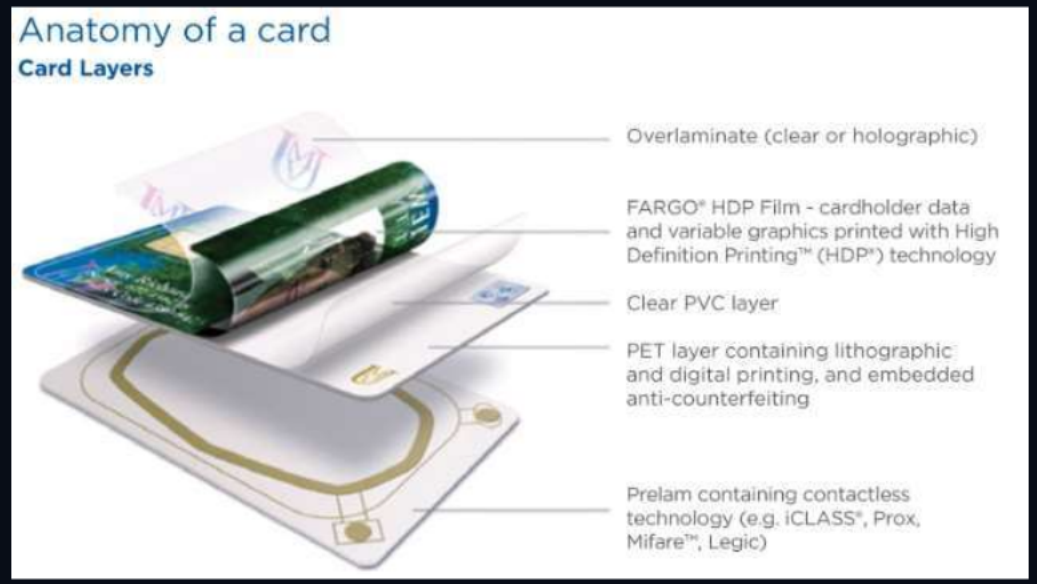
Lecture materials: PVC, polyester, polycarbonate and composites

Egypt

The lectures do not identify the exact polymer used in this Egyptian model.

Germany

Polycarbonate forms the German card body and supports integrated personalisation.



General card-layer example from Lecture 2

Lecture principle: the card body and printed data should last for the document's service life.

[L2] pp. 15, 20–26 [F]

Security background printing

Egypt





Rainbow printing creates colour transitions. Fine guilloche lines help resist copying.

Germany





Multicoloured guilloches and complex patterns protect the background against copying.

Prepress prepares artwork and colour separations. Accurate registration keeps fine details aligned.

The prepress lectures explain the process. They do not establish the exact press used for either card.

[L1] pp. 13, 18 [P1] pp. 7–13 [P2] pp. 30–32 [B] p. 2

Photo and personal data

Egypt



Multi Laser Image

The Multi Laser Image below the photo changes between the date of birth and the national emblem.

Germany



Integrated colour photo

Inkjet integrates the colour photo and document number into the material. Laser engraving adds the other personal data.

Lecture principle: personalisation should be easy to verify and difficult to reproduce.

[L1] p. 13 [L2] pp. 4, 14–16 [B] p. 2

Security when the card is tilted

Egypt



The Kinegram on the reverse shows Tutankhamun from different viewing angles.

Germany





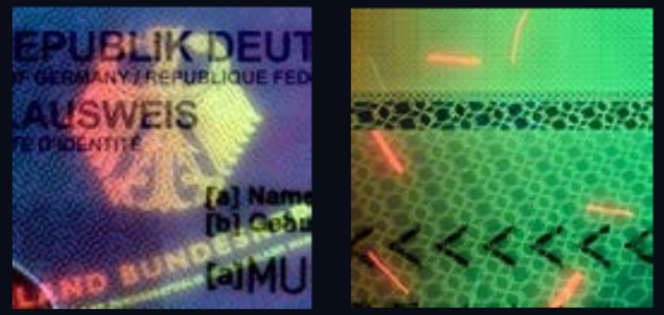
Identigram shows a holographic portrait and a 3D eagle. A separate laser image changes with the viewing angle.

Tilting the card reveals effects that a flat photocopy cannot reproduce.

[L1] p. 13 [B] p. 2

Microprinting and UV checks

Check	Egypt	Germany
Magnifying glass	Microtext and fine guilloche lines	Positive and negative microprinting
UV light	Invisible microtext and a fluorescent eagle	Fluorescent print and embedded security fibres



German UV features, from the official security guide

The lectures distinguish visible features from features that need simple checking tools.

[L1] pp. 11, 15, 18 [B] p. 2

Barcode and electronic identity

Egypt



2D barcode

A 2D barcode carries encoded data for comparison with the visible information.

Germany



eID function

A protected chip stores the facial image and fingerprints. eID supports online identification with the card and a PIN.

The clear added capability is protected electronic identification.

Lecture 1 presents a contactless chip and biometrics as future development for the Egyptian model.

[L1] pp. 15, 20–21 [B] p. 1 [D]

Comparison at a glance

Comparison	Egyptian lecture model	German 2024 model
Material	Exact polymer not specified	Polycarbonate
Background	Rainbow printing and guilloches	Multicoloured guilloches
Angle-dependent features	Multi Laser Image and Kinegram	Identigram and variable laser image
Checks with tools	Microtext and UV features	Microprinting, UV print and fibres
Digital function	2D barcode for data verification	Chip and online eID

Both use several physical security features. Germany adds protected online identification.

[L1] [L2] [B] [F] [D]

Sources

Course lectures

[L1] Security Printing, Lecture 1

Egyptian security features: pp. 11–18

Future development: pp. 20–21

[L2] Security Printing, Lecture 2

Structure and selection criteria: pp. 5–17

Card materials: pp. 20–26

[P1] Prepress Technologies, Lecture 1

Prepress and offset printing: pp. 7–13

[P2] Prepress Technologies, Lecture 2

Colour separation and registration: pp. 30–32

Additional sources

[B] German Ministry of the Interior

Security features guide, May 2024

[F] Fraunhofer IPK / Bundesdruckerei

Card manufacturing and polycarbonate

[D] Bundesdruckerei

German ID card and online eID

Dr. Rania Mohamed, 2026/2027 course lectures. Full links and page references are in the slide notes.

References and image credits are recorded in the slide notes